2019 :

Q3

১৯৯ 2020 : Q⊥

4(a) A single-layer neural network is to have six inputs and two outputs. The outputs are to be limited to and continuous over the range 0 to 1. What can you tell about the network architecture?

i) How many neurons are required?

ii) What are the dimensions of the weight matrix?

iii) What kind of transfer functions could be used?

iv) Is a bias required?

Ans: i) Two neurons, one for each output, are required.

ii) The weight matrix has two rows corresponding to the two neurons and six columns corresponding to the six inputs. The product wp is a two-element vector.

iii) Of the transfer functions we have discussed, the logsig transfer function would

be most appropriate.

iv) Not enough information is given to determine if a bias is required.



4(b) See mphil: 7(a)

4(c) See mphil: 7(b)

or

2020: 5(e)

ক(a) ১৫৫: 2020: 2(a)

5(b)

41. what is Fitness function? Discuss the associated operators used in Genetic algorithm.

- Fitness function:-

  It determines how fit an individual population is. Here, fit refers the ability of an individual to compete with other individual. It gives a fitness score to each individual. The probability that an individual will be selected for reproduction is based on its fitness score.

- Genetic operators:

  genetic operators used in GA to maintain genetic diversity. genetic diversity or variation is a necessity for evolution.

  Various operators are:

1. selection/Reproduction operator
2. crossover / Recombination
3. Mutation

1. selection :- This operator selects the best-fit individuals from the current population to be used as parents for the next generation. selection methods include roulette wheel selection, tournament selection or rank-based selection.

2. crossover: combines two/more parent solution to generate new offspring solutions. In a single-point crossover, a random point is chosen along the parent chromosomes, and the chromosomes are swapped to create two new offspring. other crossover methods include, two point crossover, uniform.

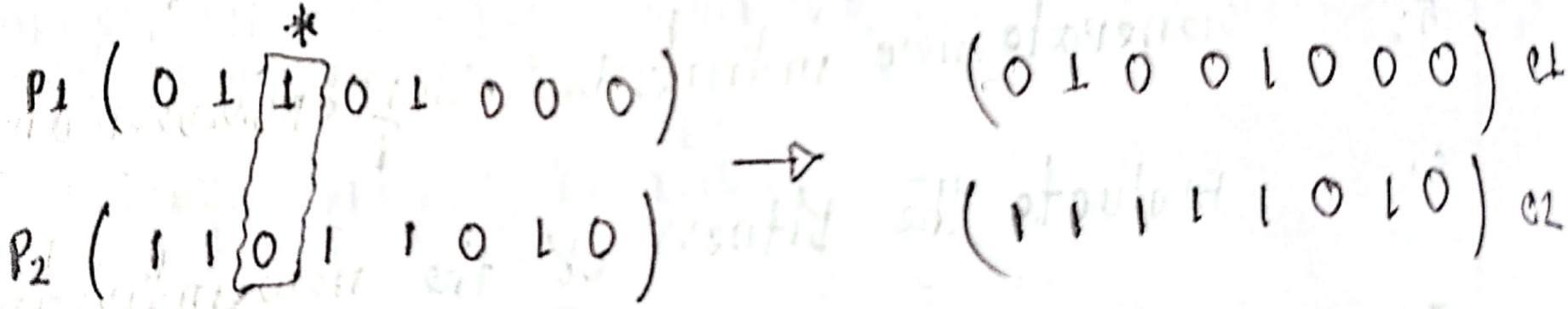
3. mutation: This operator introduces random changes to the offspring solutions to maintain genetic diversity in the population. mutation can involve flipping or changing individual bits in the chromosome, swapping or modifying genes, or adding or deleting genes.

mutation:

Before: 1 0 1 1 0 1 1 0

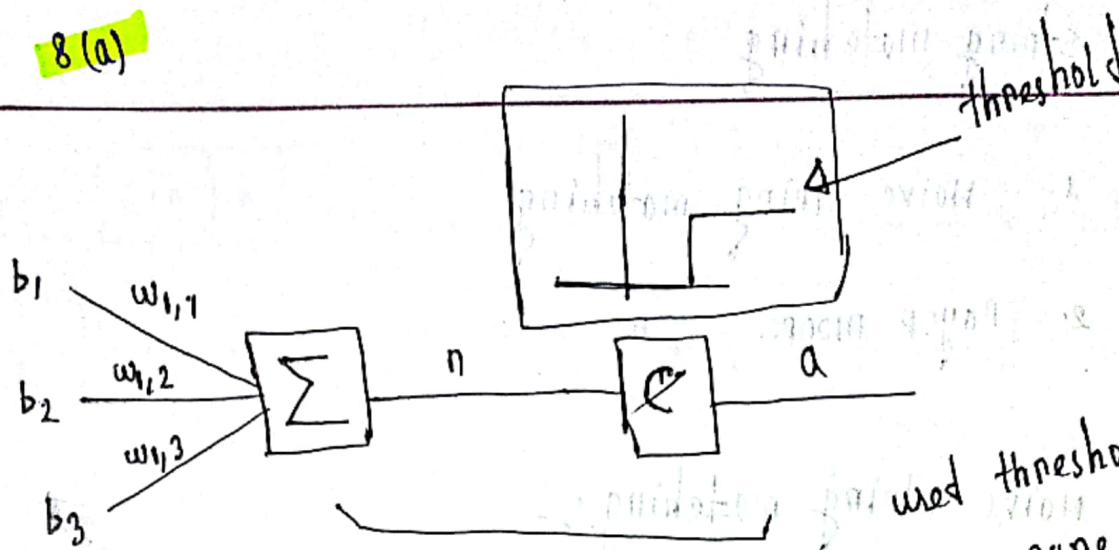
After: 0 1 1 0 0 1 1 0

crossover:



ಸಿ (c) see: 2020: 2(c)

2019 8(a)



used threshold to compare :-

a = ~~compet~~(n)

8(b)

$w_{new} = w_{old} + \text{learning rate}\left(\frac{1-(-1)}{2}\right) p_i = w_{old} + 0.2\,p_i$

let, a = output

$a = -0.5\times1 + (-1)\times0.7 + 1\times(-0.2) + 1\times0.1 + (-1)\times0.9 = -2.2 < 0 = -1$

$w = [0.5 \quad \overset{0}{\cancel{-0.7}} \quad +0.3 \quad 0.7]$ (initial update)

~~w~~ now, $a = -0.5 - 0.5 + 0 + 0.3 - 0.7 = -1$

$w = [0.3 \quad 0.2 \quad 0.5 \quad 0.5]$ (train 1)

now, a = -1

so, $w = [0.1 \quad 0.3 \quad 0.7 \quad 0.3]$ (train 2)

now, a = 1 (desired output)

2019 → 2a

Yes. GA can be used in design..... parameter.
It can be used ~~for~~ in fuzzy system and neural network.
There may be several issue that need to be resolved
while using ~~the~~ GA for membership fn design.
One issue is selection of appropriate fitness fn and

evaluation criteria. Fitness fn need to design to measure accuracy of fuzzy system output. And evaluation criteria also need to chosen carefully.

For exam: A fuzzy logic controls the speed of a car based on distance of obstacle. The membership parameter and fn should be optimized to maintain a safe distance of car from the obstacle. The fitness fn can be designed to measure the distance between car and obstacle.

Similarly GA can also be used to determine weight of Neural Net. Here GA is used to find the optimal weight that minimize the error between predicted output and actual output.

2(b) See: 2020: 2(a)

2(c) See: 2020: 2(b)